

Software delivery and organisational performance...



**...are PREDICTED by the
underlying Engineering Culture.**



Engineering Culture

Why collecting data on culture helps to empower engineering teams to improve software delivery and overall organisational performance.

“Culture is how employees'
hearts and stomachs feel about
Monday morning on Sunday
night”

Bill Marklein

Lewin's Equation:

Performance is highly influenced by the prevailing culture of an engineer's environment.

$$B = f(P, E)$$

Behaviour is a
function of the
Person and their
Environment



Kurt Lewin
Psychologist

Ron Westrum

Organisational Socialist

Widely experienced in complex organisations including medicine, aviation and the nuclear industry.



Pathological organisations are characterised by large amounts of fear and threat. People often hoard information or withhold it for political reasons, or distort it to make themselves look better.

Bureaucratic organisations protect departments. Those in the department want to maintain their “turf,” insist on their own rules, and generally do things by the book - their book.

Generative organisations focus on the mission. How do we accomplish our goal? Everything is subordinated to good performance, to doing what we are supposed to do.

Source: [The study of information flow: A personal journey](#)

Ron Westrum's Cultural Typologies

Pathologic:

Low information flow
(Power oriented)

Low co-operation/high
conflict

Emphasis on taking
care of leaders

Strict boundaries.

Bridging is discouraged

Responsibilities
shirked

Messengers are shot

Low creativity, failure
leads to scapegoating,
novelty crushed

Bureaucratic:

Medium information
flow
(Rule oriented)

Modest co-operation

Emphasis on rules and
regulation

Problems with silos,
bridging is tolerated

Narrow responsibilities

Messengers
tolerated/neglected

Creativity allowed,
failure leads to justice,
novelty leads to
problems

Generative:

High information flow
(Performance
oriented)

High co-operation

Emphasis on the
mission

Boundaryless
organisation, bridging
encouraged

Shared responsibilities

Speaking up is
encouraged

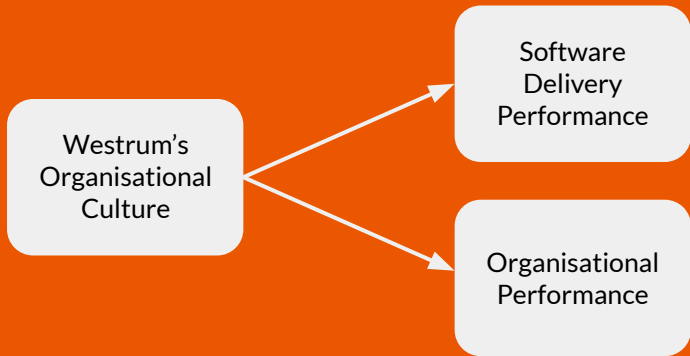
High creativity where
failure leads to inquiry
and novelty
implemented.

Toxic



Psychological
Safety

Westrum's cultural typologies...



...are **predictive** of software delivery and overall organisational performance.

Source:

Accelerate: The Science of Lean Software and DevOps, Building and Scaling High Performing Technology Organizations - Gene Kim, Jez Humble, and Nicole Forsgren

HumanCentricEngineering.com

If you are not
measuring
your
engineering
culture...



...how do you know that your
investment in software engineering
teams has the fertile ecosystem to
yield optimal returns?

Culture is qualitative so it defies easy measurement.

One approach is to survey the subjective experience of engineers, categorised by the factors of intrinsic motivation described by Deci & Ryan's Self-Determination Theory:

Mastery, Autonomy and Purpose...

Mastery

Craftsmanship

Pride in Work, Care for Quality, Technical Excellence

Autonomy

Trust

Dependability, Psychological Safety, Strong Relationships, Honesty, Vulnerability.

Purpose

Clarity

Clarity on the primary purpose of the organisation and teams. Clear Direction, Clear Career Paths, Clear Roles and Responsibilities.

Creativity

Experimentation, Prototyping, Innovation, Hypothesis-Driven Culture, Implementing Novelty.

Agency

Influence, Autonomy, Empowerment, Ownership, Accountability, Self-Organising Teams, Sustainable pace.

Leadership

Decision Making, Taking Actions, Organisational Cohesion, Collaboration, Small 'L' Leadership, Shared Responsibility.

Learning

Knowledge Sharing, Dedicated Learning Time, Communities of Practice, Continuous Improvement, Mentoring, Coaching, Growth Mindset, Feedback.

Infrastructure

Tools, Environments, Automation, Structures, Processes,, Operational Excellence

Engagement

Energy, Motivation, Flow, Contribution, Challenge, Inspiration, Wellbeing, Vitality, Connection.

Professionalism

Technical Practices, Principles, Standards, Competency, Attitudes, and Behaviours.

Information

Communication, Relationships, Alignment, Visibility, Documentation, Information Flows.

Meaning

Inclusion, Belonging, Aligned Values, Meaningful Results, Aspiration, Individual Transformation, Stability, Feeling Valued.

The **Engineering Culture Index** is an open source survey tool to measure engineer's subjective experience to benchmark culture.

See the survey questions on [GitHub](#)



Offered with humility...

The 'Culture Engineering Index' is offered in the spirit of intellectual humility and is open sourced for suggested improvements.

It doesn't promise answers, but it should prompt plenty of questions.

...open to evolution

Humans often have a bias to action.

Once we have the subjective
measurements of engineering
culture, how should we
respond to the insights?

Emergence

We must understand that culture is an emergent characteristic from the complex **ecosystem** of social interactions.

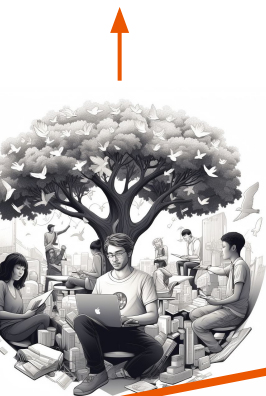
The complexity of social interactions, interdependencies and feedback loops in a software engineering ecosystem means that **we can't control culture.**

We can only **influence** the emergence of culture through a nuanced exploration of the **leverage points** within the ecosystem.



There is no silver bullet...

Engineering
Ecosystem



More pay and bonuses?
Better managers?
Restructuring?
Agile, Scrum, XP, Kanban, Lean?
More training?
Better tools?
Better engineers?
Rewrite the company values?
Copy what Google or Spotify do?

The greatest leverage point is our mindsets - our attitudes, beliefs, and the prevailing paradigms which are **unconscious** and **hard to change**.

Points of Leverage

“Folks who do systems analysis have a great belief in ‘leverage points’. These are places within a complex system (a corporation, an economy, a living body, a city, an ecosystem) where a small shift in one thing can produce big changes in everything.”



Donella Meadows
Systems Thinker

Donella Meadows

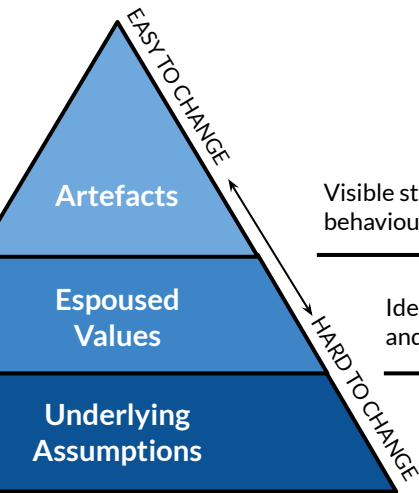
[Leverage Points: Places to Intervene in a System \(1997\)](#)

Culture Framework

Our cultural paradigms are rooted in our **unconscious mindsets** hidden from awareness and **hard to change**.



Edgar Schein
Swiss-American
Psychologist



Visible structures, processes, behaviours, events.

Ideals, goals, aspirations, discussed and agreed social norms.

Unconscious attitudes, beliefs, mental models, values. Hidden.

Culture isn't a problem to be fixed...

**...it's an emergent outcome
from our mindsets, our
conversations, our
interactions and
behaviours.**

**Patiently listening,
understanding needs,
modelling integrity and
excellence over time.**

**A rising
tide lifts all
boats...**

Questions raised by using the Engineering Culture Index:

Mastery:

- How do we create a culture of continual learning?
- How do create a compassionate feedback culture?
- How can we support engineers in taking pride in their professional and creative work?

Autonomy:

- How do we do we build organisational trust?
- How do we encourage a sense of team ownership?
- How do we optimise information flows and let engineers determine their own ways of working?

Purpose:

- How do we build a sense of belonging?
- How do we connect people to a compelling vision?
- How do we lead in a way that optimises engineer engagement?



Culture Matters

Inattention to an engineering team's subjective experience of the prevailing culture means we will miss opportunities to leverage their best performances.

Visit HumanCentricEngineering.com to learn more about services available to support the human side of building software.